Section 14.7: Maximum and Minimum Values

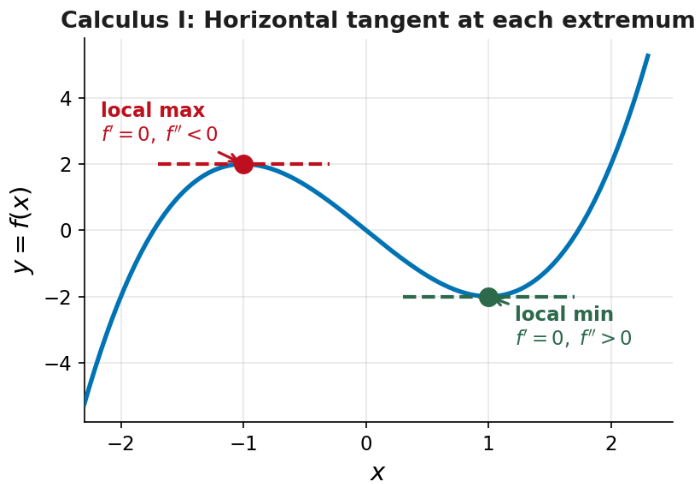
Partial Derivatives

Optimization: From Calc I to Calc III

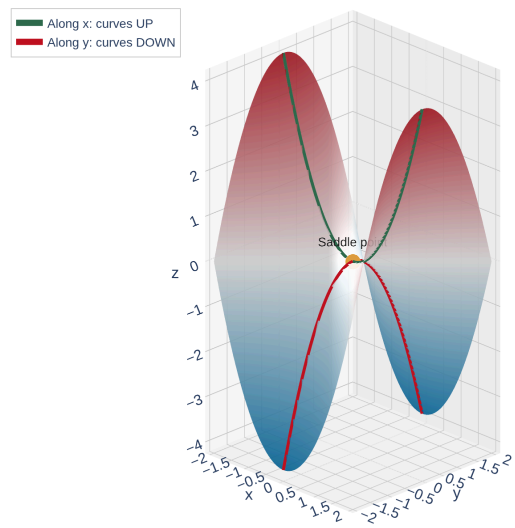
In Calculus I you learned a systematic optimization procedure. Let's see what changes in Calculus III.

	Calculus I ($y = f(x)$)	**Calculus III ($z = f(x, y)$)**
Critical points	$f'(x) = 0$ or DNE	$f_x = 0$ and $f_y = 0$ (or DNE)
Classify	2nd Derivative Test	2nd Derivative Test (with D)
New phenomenon	—	Saddle points!
Absolute extrema	Check critical pts + endpoints	Check critical pts + boundary

The big ideas carry over — but the **boundary** is now a curve instead of two endpoints, and **saddle points** are genuinely new.



Familiar: Horizontal tangent ($f' = 0$) at each local extremum. Classify with the sign of f''.



Interactive version

New in Calc III: At a **saddle point** the tangent plane is horizontal, but the surface curves *up* in one direction and *down* in another.

Local Extrema and Critical Points

Can a differentiable function f have a local maximum at (a, b) with $f_x(a, b) = 3$?

Think: what does f_x(a,b) = 3 mean geometrically?

Finding Critical Points

Critical Point

A point (a, b) is a **critical point** of f if:

- $f_x(a, b) = 0$ and $f_y(a, b) = 0$, **or**
- one (or both) partial derivatives do not exist there

Every local extremum occurs at a critical point — but not every critical point is an extremum!

Procedure for finding critical points:

1. Compute f_x and f_y
2. Set $f_x = 0$ and $f_y = 0$ simultaneously

3. Solve the system of equations for x and y

Example: Find the Critical Points

Example 1

Find the critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

Compute f_x and f_y, set both equal to zero, and solve the system.

The Second Derivative Test

Now that we can *find* critical points, how do we *classify* them?

Second Derivative Test

Suppose the second partial derivatives of f are continuous near (a, b) and $f_x(a, b) = f_y(a, b) = 0$. Define:

$$D = D(a, b) = f_{xx}(a, b) \cdot f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

Condition	Conclusion
$D > 0$ and $f_{xx} > 0$	**Local minimum**
$D > 0$ and $f_{xx} < 0$	**Local maximum**
$D < 0$	**Saddle point**
$D = 0$	**Inconclusive** (test fails)

Think of D as a measure of **curvature agreement**: when $D > 0$ the surface curves the same way in all directions. When $D < 0$ it curves **up** in one direction and **down** in another — a saddle!

Saddle Points: The New Phenomenon

A **saddle point** has no analog in single-variable calculus. At a saddle point:

- $f_x = 0$ and $f_y = 0$ (the tangent plane is horizontal)
- The surface curves **up** in one direction and **down** in another
- The point is neither a local max nor a local min

Think of a mountain pass (or the shape of a Pringles chip): it's the *highest* point on the trail going one direction, but the *lowest* point going the other direction.

$D < 0$ means the **mixed partial** f_{xy} is large relative to f_{xx} and f_{yy} — the surface is twisting.

Caution: When $D = 0$

Example 2

Show that $D = 0$ at the origin for $f(x, y) = x^4 + y^4$. Is $(0, 0)$ actually an extremum?

Compute all second partials at (0,0) and find D. Then argue from the formula for f.

Example: Classify the Critical Points

Example 3

Classify all critical points of $f(x, y) = 2x^3 + xy^2 + 5x^2 + y^2$.

(We already found them: $(0, 0)$, $(-\frac{5}{3}, 0)$, $(-1, 2)$, $(-1, -2)$.)

Compute f_xx, f_yy, f_xy, then evaluate D at each critical point.

Your Turn: Find and Classify Critical Points

Example 4

Find and classify the critical points of $f(x, y) = x^2 + y^2 - 2x - 6y + 14$.

Set f_x = 0 and f_y = 0, then compute D.

The Extreme Value Theorem

Recall from Calc I: if f is continuous on $[a, b]$, then f attains an absolute max and an absolute min on $[a, b]$.

Extreme Value Theorem (Two Variables)

If f is continuous on a **closed, bounded** set D in $\mathbb{R}^2$, then f attains an absolute maximum and an absolute minimum on D.

Closed set: includes all its boundary points (like $\leq$ vs. $<$)

Bounded set: fits inside some large disk (not stretching to infinity)

Cautionary notes:

- Absolute extrema may occur where f is **not differentiable**
- Absolute extrema may occur **on the boundary** (not just interior critical points)
- There can be infinitely many absolute extrema

Procedure for finding absolute extrema on a closed, bounded set D:

1. Find all **critical points** of f inside D
2. Find the extreme values of f **on the boundary** of D
3. The largest value from steps 1 and 2 is the absolute max; the smallest is the absolute min

Example: Absolute Extrema on a Rectangle

Example 6

Find the absolute maximum and minimum values of

$$f(x,y) = x^4 + y^4 - 4xy + 2$$

on the set $D = \{(x,y) \mid 0 \le x \le 3,\ 0 \le y \le 2\}$.

Step 1: Interior critical points. Set $f_x = 0$ and $f_y = 0$:

$$f_x = 4x^3 - 4y = 0 \ \Rightarrow\ y = x^3$$

$$f_y = 4y^3 - 4x = 0 \ \Rightarrow\ x = y^3$$

Substitute y = x\^{}3 into x = y\^{}3 to find the interior critical points in D.

Checking the Boundary (Continued)

Step 2: Check f on each edge of the rectangle $[0,3] \times [0,2]$.

Evaluate f on all four edges. On each edge, one variable is fixed.

Application: Minimum Surface Area Box

Example 7

Find the dimensions of a rectangular box with volume $1000\ \mathrm{cm}^3$ that has the smallest possible surface area.

Setup: Let the box have dimensions x, y, z (in cm).

- **Constraint:** $xyz = 1000 \Rightarrow z = \frac{1000}{xy}$
- **Minimize:** $S = 2xy + 2xz + 2yz$

Substitute z = 1000/(xy) into S, then find critical points.

Homework

Section 14.7: 1, 3, 5-21 odd, 25, 33, 37, 41, 43, 45, 47, 49, 51, 55